School of Computer Science Engineering and Technology

Lab: 2

Course	B. Tech.
Type	Elective
Course Code	CSET346
Course Name	NATURAL LANGUAGE PROCESSING
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CO-Mapping

Question	CO1	CO2	CO3
Q1	√		
Q2		√	
Q3		√	√
Q4			√

NLP Lab – WordNet, Semantic Relationships and Contextual Meaning

Objective:

This lab aims to introduce students to lexical semantics using WordNet. Students will learn to identify synsets and hypernyms, calculate semantic similarity between words, and analyze how homonyms and contextual words can have different meanings depending on their usage.

Problem Statement

Words in natural language may have multiple meanings and semantic relationships. Your task is to use WordNet and suitable Python libraries to explore word senses, hypernym relationships, semantic similarity, homonyms, and the role of context in determining word meaning.

Datasets / Resources

- NLTK WordNet Corpus – used for synsets, definitions, examples, hypernyms, and semantic similarity.
- NLTK Brown Corpus – used to obtain real-world contextual sentences.
- A small manually created sentence collection containing homonyms and contextual words.
- NLTK package and Python libraries such as pandas for analysis and tabular output.

Instructions

- Use Python and the NLTK library for all WordNet-related experiments.
- Download the required NLTK resources before executing the code.
- Display intermediate and final outputs for each experiment.
- Use suitable comments to explain the major steps in your code.
- Use the provided solutions as reference implementations and understand the output before submission.

Q1. Use NLTK WordNet to explore synsets and meanings of selected English words.

(a) Download and load the WordNet corpus using NLTK.

(b) Select at least 10 English words such as car, bank, bat, dog, light, plant, book, teacher, student, and computer.

(c) Display all or selected synsets for each word.

(d) Display the synset name, definition, and example sentence where available.

(e) Identify at least three words having multiple synsets and briefly discuss their different meanings.

Expected Output:

Word list, synset names, definitions, example sentences, and observations about multiple meanings.

Q2. Use WordNet to identify hypernyms and analyze hierarchical semantic relationships.

(a) Select at least 10 nouns from WordNet.

(b) Identify a suitable synset for each word.

(c) Extract its direct hypernyms.

(d) For at least five words, trace the hypernym hierarchy for two or more levels.

(e) Display the word, synset, and hypernym chain in a Pandas DataFrame.

Expected Output:

A table showing words, synsets, direct hypernyms, and hierarchical semantic relationships.

Q3. Calculate WordNet semantic similarity between different word pairs.

(a) Select at least eight word pairs, including semantically similar and dissimilar words.

(b) Obtain appropriate WordNet synsets for each word.

(c) Calculate WordNet path similarity for each pair.

(d) Compare pairs such as car–automobile, dog–cat, dog–animal, and dog–computer.

(e) Store the results in a Pandas DataFrame and rank the pairs according to similarity score.

(f) Briefly explain why different pairs receive different similarity values.

Expected Output:

Word pairs, selected synsets, similarity scores, and a ranked table with observations.

Q4. Analyze homonyms and contextual words to demonstrate the importance of context in determining meaning.

(a) Select at least five homonyms such as bank, bat, bark, light, and match.

(b) Create at least two sentences for each word, where the word has different meanings.

(c) Use WordNet to display different synsets corresponding to the homonyms.

(d) Compare the contextual sentences and manually identify the intended meaning.

(e) For at least three contextual examples, explain which surrounding words help determine the correct meaning.

(f) Create a final table containing the word, sentence, intended meaning, and relevant contextual clue.

Expected Output:

Contextual sentences, WordNet synsets, intended meanings, contextual clues, and a final comparison table.